Pacific-Basin Finance Journal 92 (2025) 102819



Contents lists available at ScienceDirect

Pacific-Basin Finance Journal

journal homepage: www.elsevier.com/locate/pacfin





The impact of ESG ratings on firm risks in Taiwan's market☆

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ARTICLE INFO

JEL classification:
G14
G30
G32
M14
M40

Keywords:
Crash risk
ESG
Financial constraint
Financial risk

ABSTRACT

This study investigates the impact of a firm's ESG achievements on various dimensions of firm risk, including financial risks, stock price crash risks, and financial constraints. Empirical evidence reveals a negative association between ESG scores and financial risks, as well as financial constraints, indicating that higher ESG performance contributes to lower overall firm risk. The reduction in crash risks becomes particularly evident during the COVID-19 crisis, highlighting the protective role of ESG engagement in times of market stress. These findings suggest that strategic investment in ESG activities enhances a firm's resilience and stability during both normal and adverse conditions.

1. Introduction

Environmental, Social, and Governance (ESG) considerations have asserted their centrality in contemporary business activities. As global sustainable investment reached $35.3 trillion at the start of 2020 in five major markets, a 15 % increase in two years (Global Sustainable Investment Alliance, 2021), Social Responsibility Investment (SRI) investors, fund managers, and corporate executives commonly ask whether ESG practices establish positive rewards for a firm's performance and risk management. Since then, the relationship between ESG engagement and a firm's financial performance has been widely explored in the literature.[1] However, there is no conclusive evidence regarding the relationship between ESG diligence and firm risks. For example, studies such as Cai et al. (2016), El Ghoul et al. (2011), Goss and Roberts (2011), Khorilov and Kim (2024), Kim et al. (2014), and Jo and Na (2012) show that ethical business practices enable firms to hedge against financial risks and stock price crash risks and alleviate financial constraints. In

☆ The authors appreciate helpful comments and suggestions from Konan Chan, Hong-Ren Huang, Ching-Hung Chang, Meng-Lan Yueh, as well as participants at the 2024 FeAT International Conference and 2024 International Conference of Taiwan Finance Association. This paper is partly drawn from Shun-Fen Chang's research thesis entitled "Does Sustainability Report Matter? The Impact of ESG Ratings on Firm Risks in Taiwan's Market".

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[1] See Renneboog et al. (2008), Servaes and Tamayo (2013), Dimson et al. (2015), Dyck et al. (2019), Hartzmark and Sussman (2019), and Ng and Rezaee (2020).

https://doi.org/10.1016/j.pacfin.2025.102819
Received 28 August 2024; Received in revised form 15 May 2025; Accepted 16 May 2025
Available online 19 May 2025
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contrast, McWilliams et al. (2006), Hemingway and Maclagan (2004), Kim (2010), and Humphrey et al. (2012) document that ESG and corporate social responsibility (CSR) initiatives may lead to agency problems, reduced transparency in financial reporting, or may be irrelevant to firm risks. This study, therefore, aims to conduct a comprehensive analysis of the relationship between ESG engagement and firm risks to provide a deeper understanding of its impact.

This study focuses on Taiwan's market, which presents a compelling opportunity for analysis due to its well-established sustainability disclosure framework. In 2014, the Taiwan Stock Exchange (TWSE) introduced *Taiwan Stock Exchange Corporation Rules Governing the Preparation and Filing of Sustainability Reports by TWSE Listed Companies*, a set of guidelines that significantly shape a framework for ESG reporting and encourage listed companies to disclose ESG information. Since then, an increasing number of companies have adopted more detailed ESG disclosures, with reporting frameworks becoming progressively more sophisticated. According to a survey conducted by KPMG International (2022), Taiwan ranks among the top countries in terms of corporate sustainability reporting, with disclosure rates exceeding 90 %.[2] Moreover, Taiwan is among the world's top 10 markets in terms of the percentage of companies integrating sustainability information into their annual financial reports, obtaining third-party assurance on sustainability reports, and adhering to Task Force on Climate-related Financial Disclosures (TCFD) recommendations. Additionally, Taiwan exhibits the highest percentage of companies with sustainability representation at the Board level. These achievements highlight Taiwan's strong commitment to sustainability reporting, which provides a solid foundation for disclosure-based ESG ratings and enhances the credibility of ESG scores. Covering all listed and over-the-counter (OTC) companies from 2015 to 2021, including firms with varying ESG scores, Taiwan provides a comprehensive and inclusive setting that is well-suited for academic research into the relationship between ESG diligence and firm risks.

This study, therefore, investigates whether ESG engagement in Taiwan provides insurance-like protection, consistent with the idiosyncratic risk channel (Godfrey et al., 2009; Jo and Na, 2012), the valuation channel (El Ghoul et al., 2011; Giese et al., 2019), stakeholder theory (Kim et al., 2014), and both better stakeholder management and transparency (Cheng et al., 2014). Specifically, we examine the impact of ESG performance—measured using TEJ ESG scores—on three categories of risk: financial risks, stock price crash risks, and financial constraints.[3] Moreover, prior studies (e.g., Bouslah et al., 2018) suggest that strong ESG performance helped reduce volatility during the 2008–2009 financial crisis. Building on this idea, we, therefore, examine whether investors perceive ESG performance as a signal of resilience during crises, particularly during the COVID-19 pandemic.

This paper presents two key findings that enhance our understanding of ESG engagement in corporate risk management. First, the analysis shows a significant negative association between ESG scores and various financial risks, including idiosyncratic risk, systematic risk, and total risk. Additionally, higher ESG scores are associated with reduced financial constraints, indicating that firms with strong ESG performance are better equipped to manage their financial liabilities and have more favorable access to capital. This underscores the value of ESG practices not only in mitigating specific financial risks but also in strengthening overall financial stability. Second, while ESG scores do not directly influence stock price crash risk under normal conditions, they do play a crucial role in mitigating the impact of price crashes during periods of severe market stress, such as the COVID-19 pandemic. This finding suggests that ESG engagement may serve as a form of insurance during crises, providing firms with a buffer against extreme volatility and downturns in unprecedented circumstances. These findings highlight the strategic value of ESG initiatives in safeguarding firms against both financial risks and extraordinary market disruptions, reinforcing the role of ESG in promoting long-term corporate resilience.

This study contributes to the literature on ESG diligence in several aspects. First, this study examines a comprehensive dataset that covers over 90 % of listed companies in Taiwan's stock market. This is important because many previous studies on ESG focus only on a small subset of firms, often those with strong ESG practices, which can lead to a selection bias and limit the applicability to the broader market. By analyzing a nearly complete dataset, this study mitigates such bias and provides more representative evidence of the relationship between ESG performance and firm risks. Furthermore, Taiwan's ESG disclosure framework follows the TCFD recommendations and requires third-party assurance, enhancing data reliability. As it captures the impact of ESG engagement across a diverse range of firms, this study offers more generalizable conclusions, helping to address the debate on the impact of ESG on firm risks.

Second, this study conducts a comprehensive analysis of financial risks by examining total risk, stock price crash risk, and financial constraints, allowing for a more nuanced assessment of ESG's impact on corporate stability. By decomposing total risk, the study identifies how ESG engagement influences both systematic and idiosyncratic risks, offering insights beyond aggregate risk measures. The inclusion of stock price crash risk is particularly important, as it assesses whether ESG performance helps firms mitigate severe market downturns and extreme volatility. Additionally, analyzing financial constraints provides evidence of how ESG practices enhance access to capital and reduce financing challenges. By examining various dimensions of financial risk, this study provides a comprehensive view of how ESG contributes to mitigating corporate vulnerabilities and enhancing long-term resilience.

Third, this study provides strong evidence for the role of non-financial disclosure and ESG engagement in mitigating financial risks and reducing financial constraints. In line with Diamond and Verrecchia's (1991) framework, greater public disclosure reduces

[2] The Survey of Sustainability Reporting has been conducted by KPMG since 1993, with the 2022 survey marking its 12th edition. It examines sustainability reporting trends worldwide and is based on data from two distinct research samples: the G250 and the N100. The G250 includes the world's 250 largest companies by revenue as defined in the 2021 Fortune 500 ranking. The N100 comprises a global sample of 5800 companies, representing the top 100 companies by revenue in each of the 58 markets. The data mentioned in this section is based on the N100 sample.

[3] In Taiwan, sustainability reports from July 1, 2021, to June 30, 2022, show a 99 % adoption of the Global Reporting Initiative (GRI) among listed firms. Consequently, Taiwan Economic Journal (TEJ) ESG scores are primarily based on the GRI framework for consistency.

information asymmetry, enabling investors to make more informed decisions, thereby enhancing market confidence and stability. Increased transparency not only attracts institutional investors and improves stock liquidity but also lowers a firm's cost of capital by reducing perceived risk. Furthermore, consistent with the findings of Jo and Na (2012), Bouslah et al. (2013), and Kim et al. (2014), strong ESG performance enhances creditworthiness and access to financing, helping firms alleviate financial constraints. By fostering investor trust, strengthening corporate credibility, and improving financial resilience, ESG disclosure serves as both a risk management tool and a key mechanism for ensuring long-term financial stability.

Fourth, this study provides valuable insights for policymakers, emphasizing that ESG plays a crucial role not only in improving operational performance but also in enhancing corporate risk management. The findings highlight how transparent ESG disclosure and strong sustainability practices contribute to risk reduction, making firms more resilient to financial shocks. This study encourages better ESG integration in corporate strategies to promote long-term stability.

The remainder of this paper is organized as follows. Section 2 provides a comprehensive review of the relevant literature and develops the hypotheses. Section 3 describes the dataset, variable definitions, and summary statistics. Section 4 presents empirical findings corresponding to the hypotheses. Section 5 discusses robustness checks and addresses potential endogeneity concerns. Finally, Section 6 concludes the study with key findings and implications.

2. Literature review and hypothesis development

2.1. ESG and financial risks

ESG achievements serve as protective shields for firms against significant risk exposure, such as idiosyncratic risk, systematic risk, and total risk, through both the idiosyncratic risk channel and the valuation channel. The idiosyncratic risk channel, according to Godfrey et al. (2009), Jo and Na (2012), Oikonomou et al. (2012), and Giese et al. (2019), posits that companies with strong ESG engagement typically adhere to high-risk control standards across their operations and supply chain management. The high standards reduce the likelihood of severe incidents such as fraud, embezzlement, corruption, or litigation cases (Hong and Kacperczyk, 2009). This mechanism is often referred to as "insurance-like protection" that reduces corporate risk exposure (Godfrey et al., 2009). Empirical findings show a negative association between ESG performance and idiosyncratic risk (Boutin-Dufresne and Savaria, 2004; Lee and Faff, 2009; Bouslah et al., 2013; Benlemlih et al., 2018; Giese et al., 2019). However, contrasting perspectives also exist. Becchetti et al. (2015) argue that ESG involvement may increase a firm's idiosyncratic risk by reducing its flexibility in responding to adverse productivity shocks. Humphrey et al. (2012) analyze CSP portfolios instead of directly addressing idiosyncratic risk in individual firms in the UK and find no significant difference in the aggregate unsystematic risk across firms.

The valuation channel, as demonstrated by El Ghoul et al. (2011) and Giese et al. (2019), suggests that companies with robust ESG practices are less vulnerable to systematic market shocks, leading to lower systematic risk. A reduction in systematic risk implies that investors require a lower rate of return (Benabou and Tirole, 2010), reducing the company's cost of capital and enhancing its valuation. Empirical studies by Jo and Na (2012), Oikonomou et al. (2012), and Albuquerque et al. (2019) support the hypothesis that ESG engagement reduces systematic risk, while Salama et al. (2011) find a mildly inverse relationship between environmental performance and systematic financial risk in the UK. Benlemlih et al. (2018) agree that ESG can improve a firm's valuation, although they do not observe a direct link between environmental and social disclosures and systematic risk. As idiosyncratic and systematic risks decline, total risk is also expected to decrease. Studies find that CSR/ESG engagement inversely affects stock volatility (Jo and Na, 2012; Bouslah et al., 2013; Benlemlih et al., 2018; Broadstock et al., 2021).

2.2. ESG and stock price crash risks

The relationship between ESG initiatives and stock price crash risk is complex and debated, involving stakeholder theory and the agency perspective. Stakeholder theory suggests that higher ESG initiatives enhance information transparency, which reduces the likelihood of hiding bad news and, consequently, lowers the probability of stock price crash risk (Kim et al., 2014; Feng et al., 2022). Conversely, the agency perspective argues that ESG can trigger a "whitewash" mechanism, enabling managers to hoard bad news, signal decent images, and mislead investors, hence potentially leading to severe stock price crash risks (Kim et al., 2014; Feng et al., 2022). Empirical evidence shows that ESG mitigates crash risk, supporting the stakeholder view (Kim et al., 2014; Hoepner et al., 2024; Bae et al., 2021; Yu et al., 2023), while Chen et al. (2001) find conflicting results. In contrast, supporting the agency perspective, Ben-Nasra and Ghouma (2018) demonstrate that while managers may attempt to cover up bad news by enhancing employee welfare, excessive employee welfare expenditures will increase stock price crash risk.

2.3. ESG and financial constraints

Firms with higher ESG scores tend to experience lower systematic risk and higher valuations because of reduced information asymmetry and a larger pool of investors, lowering the cost of capital (Hong and Kacperczyk, 2009; El Ghoul et al., 2011). Many studies show that firms with CSR initiatives, environmental risk management, higher non-financial disclosure, or better ESG ratings have cheaper financing costs (Heal, 2005; Sharfman and Fernando, 2008; Orens et al., 2010; El Ghoul et al., 2011; Dhaliwal et al., 2011) and that firms with social or environmental concerns increase financing costs (Goss and Roberts, 2011; Chava, 2014). However, ESG implementation requires capital investment, which would constrain capital allocation to other business tasks (Chan et al., 2017). Nazir et al. (2022) find that ESG performance increases the cost of capital, as investors perceive ESG initiatives as additional expenditures.

In addition to the cost of capital, high-ESG companies face fewer financial constraints, assessed by the KZ, SA, and WW indices, because of greater transparency, better stakeholder management, and lower agency costs (Cheng et al., 2014). Hong et al. (2012), Chan et al. (2017), and Bae et al. (2021) also demonstrate that strong corporate social performance helps alleviate financial constraints. In contrast, Zhang (2022) explores the relationship between ESG and financial constraints from an agency perspective and finds that highly leveraged firms and those facing financial difficulties are more prone to greenwashing.

2.4. ESG, firm risks, and the COVID-19 crisis

Most entrepreneurs prioritize risk and opportunity assessment when managing their businesses, considering ESG factors only when time or resources permit. However, previous studies show that adopting an ESG-first approach can be beneficial, particularly due to the insurance-like function of ESG in challenging times. Bouslah et al. (2018) find that corporate social performance significantly reduces volatility in the US during the 2008–2009 financial crisis. Chiaramonte et al. (2022) demonstrate that higher ESG scores reduce EU bank fragility during tough times and emphasize that longer-term ESG disclosures offer greater stability benefits during financial turmoil. Broadstock et al. (2021), using a sample of China CSI300 listed companies, show that ESG performance plays an enhanced role during the COVID-19 pandemic by reducing total risk and increasing stock return.

2.5. Hypothesis development

Beyond prior literature, companies may also instinctively employ ESG practices to mitigate risks, especially as global discussions on ESG regulations intensify. For instance, businesses closely engaged with the EU are proactively addressing the regulatory risk of carbon taxation on high-carbon-emitting products after 2026. To avoid significant carbon taxes, these companies are investing in process and equipment improvements to reduce their carbon footprint or emissions. Such actions have direct financial implications, influencing asset pricing and firm risk. This risk management aligns with the Financial Stability Board's promotion of TCFD disclosure (Financial Stability Board, 2021).

Taken together, because of the mixed and inconclusive findings in prior literature, we aim to investigate whether ESG practices provide insurance-like protection in Taiwan's market. Firms with strong ESG engagement tend to implement higher risk control standards (Godfrey et al., 2009; Jo and Na, 2012), exhibit lower sensitivity to systematic market shocks (El Ghoul et al., 2011; Giese et al., 2019), promote greater information transparency (Kim et al., 2014), and foster stronger stakeholder relationships (Cheng et al., 2014). These mechanisms may reduce idiosyncratic risk, systematic risk, stock price crash risk, or financial constraints. This leads us to the first hypothesis:

H1. ESG ratings in Taiwan negatively contribute to financial risks, stock price crash risks, and financial constraints.

Additionally, we are interested in exploring whether firms in Taiwan can bolster their resilience during market crises through ESG-driven risk management mechanisms (Bouslah et al., 2018). Our research concentrates on the years 2020 and 2021, selected due to the initial COVID-19 shock to the Taiwan stock market in 2020 and 2021. This leads us to the second hypothesis:

H2. Firms in Taiwan with higher ESG ratings experience lower financial risks, stock price crash risks, and financial constraints during the waves of COVID-19.

3. Data and summary statistics

This study focuses on firms listed on the TSE and OTC from 2015 to 2021. ESG scores, stock trading data, and accounting figures are obtained from the Taiwan Economic Journal (TEJ) database. The sample includes firms listed on the TSE and OTC, excluding financial institutions. Firms must have a positive book value of equity and at least 200 daily stock returns each year to be included in the analysis. Observations with missing values are excluded from the sample. Control variables include leverage ratio (LEV), return on assets (ROA), market-to-book ratio (MB), market capitalization (MKTCAP), change in trading volume (DTURNOVER), and past returns (RET) and are defined in Appendix A.

Taiwanese public companies are encouraged to submit sustainability reports annually, which are assessed by the Corporate Governance Evaluation conducted by TWSE. As a result, most companies comply, allowing TEJ to gather and compile data that covers over 90 % of listed companies in Taiwan. TEJ integrates unique datasets from multiple sources and consolidates this information into ESG ratings. Specifically, the TEJ ESG dataset consists of three pillars- environmental (E), social (S), and governance (G). Scores in each category range from 0 to 100. In this study, we extract the pillar scores (E, S, and G scores) and the aggregate ESG score as independent variables for our analysis.

Understanding the inherent risks associated with firms requires examining both broad risk categories and specific financial metrics, including financial risks, stock price crash risks, and financial constraints. Detailed definitions are provided in Appendix A. To measure financial risks, we employ total risk, systematic risk, and idiosyncratic risk. Total risk (SIGMA), adapted from Ang et al. (2006) and Jo and Na (2012), provides an assessment of the overall uncertainties a firm faces. Systematic risk (BETA), as discussed in Sharpe (1964), Lintner (1965), and Mossin (1966), captures market-wide disturbances that can significantly affect firm value. Idiosyncratic risk (IVOL_FF3), following the framework of Ang et al. (2006), offers a clearer view of firm-specific uncertainties. Higher values of SIGMA, BETA, and IVOL_FF3 indicate greater financial risk.

To evaluate stock price crash risks, following Chen et al. (2001) and Kim et al. (2014), we employ the negative coefficient of

skewness (NCSKEW) and down-to-up volatility (DUVOL). Higher values of NCSKEW and DUVOL indicate a greater likelihood of stock price crash risk. Proxies for financial constraints include the KZ, WW, and SA indices. The KZ index is adapted from Kaplan and Zingales (1997), Baker et al. (2003), and Cheng et al. (2014). The WW index is calculated according to Whited and Wu (2006) and Cheng et al. (2014). The SA index is estimated based on Hadlock and Pierce (2010) and Cheng et al. (2014). For financially constrained firms, the values of the KZ, WW, and SA indices tend to be higher.

Table 1 presents descriptive statistics for the sample, including ESG scores, firm risks, and control variables. All variables, except ESG scores and financial constraint indices, are winsorized at the 1 % and 99 % levels to mitigate the influence of extreme outliers and ensure consistency in the analysis. The mean values for ESG, ESG_E, ESG_S, and ESG_G are 54.62, 54.70, 54.72, and 54.50, respectively. The corresponding standard deviations are 7.86 (ESG), 11.05 (E), 10.45 (S), and 10.95 (G), indicating that firms may differ more significantly in their environmental and governance practices than in their aggregate ESG engagement. The higher dispersion in E and G components may reflect differences in industry-specific regulatory pressures, environmental exposure, or the degree of institutional commitment to governance practices.

4. Empirical results

4.1. ESG and firm risks

To investigate the effect of ESG achievements on firm risks, we conduct regressions with the fixed effects as follows.

$$\begin{aligned} Firm\ Risk_{i,t} = {} & \alpha + \beta_1 ESG_{i,t-1} + \beta_2 LEV_{i,t-1} + \beta_3 ROA_{i,t-1} + \beta_4 MB_{i,t-1} + \beta_5 Ln(MKTCAP)_{i,t-1} + \beta_6 DTURNOVER_{i,t-1} + \beta_7 RET_{i,t-1} \\ & + \beta_8 SIGMA_{i,t-1} + Industry_Dummies + Year_Dummies + \varepsilon_{i,t} \end{aligned} \quad (1)$$

where the dependent variable is one of the firm risk measures, financial risks, crash risks, and financial constraints, of firm *i* in year *t*. The key independent variable is the ESG score of firm *i* in year *t*-1. Control variables include leverage ratio, return on assets, market-to-book ratio, market capitalization, change in trading volume, prior return, and total risk. Year and industry fixed effects are included, and standard errors are clustered by industry and year to account for heteroskedasticity.

The empirical results for the effect of ESG on firm risks are presented in Table 2. In terms of financial risks, columns (1) to (3) include total risk, systematic risk, and idiosyncratic risk as the dependent variable, respectively. Across all three measures, the coefficients for ESG are negative and statistically significant at the 1 % level. The results support our first hypothesis that ESG engagement reduces financial risks.

In columns (4) and (5), we evaluate stock price crash risk using the negative coefficient of skewness and down-to-up volatility. The coefficients for ESG are insignificant. This outcome may be influenced by the presence of Taiwan's ± 10 % daily price limit rule, which can suppress extreme downside movements and delay full price adjustments (Kim and Rhee, 1997). This price limit mechanism may reduce the observability of crash events, weakening the predictive power of ESG scores in this context. We address this issue further by focusing on crisis periods in Section 4.3.

Columns (6) to (8) show the results for financial constraints using the KZ, WW, and SA indices. The results show that ESG is negatively associated with both the KZ and WW indices, suggesting that firms with better ESG performance will face fewer financial constraints. The SA index, however, yields an insignificant coefficient. This may be due to the index's high sensitivity to firm size. As

Table 1
Summary statistics.

Variable	N	Mean	STD	Min	Q1	Median	Q3	Max
ESG	10,966	54.620	7.858	29.890	49.100	54.030	59.740	83.730
ESG_E	10,966	54.700	11.046	25.350	45.940	52.380	62.110	90.960
ESG_S	10,966	54.722	10.451	26.580	46.960	53.780	61.710	91.000
ESG_G	10,966	54.503	10.947	19.650	47.150	55.080	62.080	84.410
SIGMA	10,966	2.166	0.920	0.599	1.455	2.056	2.748	4.775
BETA	10,966	0.756	0.438	−0.121	0.419	0.714	1.055	1.892
IVOL_FF3	10,966	1.930	0.844	0.558	1.292	1.817	2.419	4.510
NCSKEW	10,966	−0.705	0.698	−2.854	−1.094	−0.654	−0.259	1.068
DUVOL	10,966	−0.584	0.462	−1.654	−0.907	−0.598	−0.274	0.573
KZ_Index	10,966	−0.464	1.963	−8.543	−1.454	−0.038	0.919	3.317
WW_Index	10,966	−0.724	0.082	−0.970	−0.774	−0.726	−0.674	−0.484
SA_Index	10,966	−2.315	0.937	−4.805	−2.947	−2.447	−1.846	2.067
LEV	10,966	0.245	0.204	0.000	0.047	0.225	0.393	0.767
ROA	10,966	0.028	0.081	−0.314	<0.000	0.033	0.070	0.221
MB	10,966	1.911	1.535	0.472	0.982	1.449	2.234	9.631
Ln(MKTCAP)	10,966	8.275	1.404	5.617	7.277	8.102	9.098	12.635
DTURNOVER	10,966	0.061	0.852	−2.739	−0.179	−0.007	0.164	4.091
RET	10,966	0.056	0.145	−0.269	−0.033	0.038	0.127	0.539

This table presents the descriptive statistics for ESG scores, risk measures, and all control variables of 10,966 firm-year observations from 2015 to 2021. Variables are defined in Section 2 and Appendix A. Variables except ESG scores and financial constraint indices are winsorized at levels 1 % and 99 %. For measures of financial constraints, we winsorize the components of indices before calculating them.

Table 2
The effect of ESG on firm risks.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	SIGMA	BETA	IVOL_FF3	NCSKEW	DUVOL	KZ_Index	WW_Index	SA_Index
Intercept	2.9056***	−0.0244	3.0048***	−1.5850***	−0.8624***	−0.5426***	−0.3416***	−7.3874***
	(19.27)	(−0.20)	(22.34)	(−15.58)	(−12.57)	(−3.02)	(−53.60)	(−67.62)
ESG	−0.0064***	−0.0031***	−0.0056***	0.0004	0.0005	−0.0136***	−0.0003***	0.0010
	(−4.90)	(−3.90)	(−5.14)	(0.46)	(0.97)	(−6.59)	(−6.68)	(1.10)
LEV	0.3370***	0.0870***	0.3091***	0.0440	−0.0137	4.3331***	−0.0569***	0.7280***
	(7.18)	(3.18)	(6.47)	(1.00)	(−0.42)	(54.86)	(−19.51)	(19.97)
ROA	−3.6220***	−0.0357	−3.5931***	−0.1376	−0.1016	−10.6604**	−0.1721***	−0.1377
	(−21.43)	(−0.52)	(−21.67)	(−1.01)	(−0.94)	(−38.32)	(−18.13)	(−0.95)
MB	0.1363***	−0.0201***	0.1531***	0.0634***	0.0548***	−0.2850***	0.0131***	−0.1343***
	(11.37)	(−5.46)	(15.18)	(8.76)	(9.35)	(−15.77)	(31.21)	(−15.88)
Ln(MKTCAP)	−0.1135***	0.0972***	−0.1514***	0.0443***	0.0107	0.0492***	−0.0476***	0.5900***
	(−10.15)	(11.68)	(−14.44)	(4.51)	(1.51)	(4.85)	(−115.69)	(42.33)
DTURNOVER	0.0638***	0.0207	0.0541***	0.0070	0.0241*	−0.0034	−0.0004	−0.0257**
	(2.86)	(1.12)	(3.34)	(0.47)	(1.90)	(−0.09)	(−0.46)	(−2.20)
RET	1.0705***	0.5029***	0.8035***	0.4252***	0.3210***	−0.2999	−0.0295***	−0.5286***
	(5.03)	(4.44)	(4.32)	(4.96)	(4.80)	(−1.45)	(−6.06)	(−7.97)
SIGMA	–	–	–	0.0810***	−0.0143	0.2155***	0.0105***	0.1119***
				(5.08)	(−1.28)	(9.69)	(15.38)	(7.52)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.3971	0.3309	0.3793	0.0889	0.0840	0.6244	0.8199	0.6933
Obs.	9213	9213	9213	9213	9213	9213	9213	9213

This table presents the regression analysis on the effect of ESG score on firm risks from 2015 to 2021 with the employment of cluster standard errors. For financial risks, columns (1), (2), and (3) present the results on total risk, systematic risk, and idiosyncratic risk, respectively. For stock price crash risks, columns (4) and (5) present the results on the negative coefficient of skewness and down-to-up volatility, respectively. For financial constraints, columns (6), (7), and (8) present the results on KZ, WW, and SA indices, respectively. The independent variables are the values from the previous year (*t-1*). Year and industry fixed effects are included. Numbers in parentheses are *t*-values. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels.

most of Taiwan's corporations are small or medium-sized enterprises (SMEs), ESG initiatives might be too costly for SMEs (Makni et al., 2009). This may offset the benefits of ESG-related transparency and credibility.

4.2. E, S, and G pillars and firm risks

We further examine the distinct effects of the environmental, social, and governance pillars of ESG achievements on firm risks. Specifically, we run separate regressions for each pillar score against financial risks, stock price crash risks, and financial constraints, using the same control variables and fixed effects in Eq. (1). Panel A of Table 3 presents results for the environmental pillar. The E score is negatively associated with idiosyncratic risk (IVOL_FF3) and all three financial constraint measures (KZ, WW, and SA indices). These results suggest that environmental initiatives help firms reduce firm-specific uncertainty and improve access to external financing. The effect on crash risks, however, is insignificant, indicating that environmental engagements alone may not strongly influence downside tail risks in normal market conditions.

Panel B shows results for the effect of the social pillar on firm risks. The S score is negatively related to all three measures of financial risk (SIGMA, BETA, and IVOL_FF3) as well as financial constraint indices (the KZ, WW, and SA indices) at the 1 % level. These findings highlight the effectiveness of social engagement in enhancing firm stability, reducing risk exposures, and easing financial constraints. The relationships with crash risk measures remain negative but insignificant, consistent with the earlier findings in Section 4.1.

Panel C presents results for the effect of the governance pillar on firm risks. The G score exhibits a strong negative association with financial risks and the KZ and WW indices. These findings further confirm the view that sound corporate governance contributes to reduced uncertainty and better financing conditions. However, larger firms often score higher in governance but may simultaneously exhibit higher SA index values due to their structural characteristics. This size-driven effect may offset the positive impact of governance on financial conditions, leading to an insignificant association between the G score and the SA index.

Consistent with prior literature, the E, S, and G pillars have varying degrees of influence on firm risk profiles. Social and governance dimensions show the most consistent and significant associations with reduced financial risk and financial constraints, whereas the effect of the environmental pillar is more concentrated in alleviating financing limitations. None of the pillars demonstrates a statistically significant effect on crash risk during normal periods, reflecting the earlier argument that Taiwan's market structure may hinder the detection of the potential influence of ESG on crash risk.

Table 3
The effect of E, S, and G on firm risks.

Panel A. ESG_E

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	SIGMA	BETA	IVOL_FF3	NCSKEW	DUVOL	KZ_Index	WW_Index	SA_Index
Intercept	2.7167***	−0.1194	2.8430***	−1.5540***	−0.8419***	−0.9526***	−0.3499***	−7.2321***
	(18.38)	(−1.03)	(21.42)	(−15.95)	(−12.15)	(−5.96)	(−56.54)	(−76.10)
ESG_E	−0.0013	−0.0005	−0.0013*	−0.0005	<0.0001	−0.0037***	−0.0001***	−0.0036***
	(−1.36)	(−0.91)	(−1.68)	(−0.72)	(0.02)	(−3.12)	(−4.77)	(−5.86)
LEV	0.3474***	0.0918***	0.3184***	0.0444	−0.0142	4.3524***	−0.0563***	0.7334***
	(7.63)	(3.48)	(6.82)	(1.00)	(−0.43)	(56.46)	(−19.52)	(20.49)
ROA	−3.6823***	−0.0653	−3.6455***	−0.1338	−0.0978	−10.7515**	−0.1742***	−0.1241
	(−21.57)	(−0.93)	(−21.70)	(−0.97)	(−0.90)	(−38.74)	(−18.49)	(−0.87)
MB	0.1378***	−0.0193***	0.1544***	0.0630***	0.0547***	−0.2846***	0.0131***	−0.1366***
	(11.64)	(−5.19)	(15.46)	(8.67)	(9.30)	(−15.34)	(31.35)	(−16.04)
Ln(MKTCAP)	−0.1259***	0.0907***	−0.1619***	0.0468***	0.0120*	0.0274***	−0.0479***	0.6042***
	(−11.41)	(10.47)	(−15.55)	(4.77)	(1.73)	(2.68)	(−110.99)	(40.70)
DTURNOVER	0.0631***	0.0203	0.0534***	0.0071	0.0243*	−0.0079	−0.0005	−0.0254**
	(2.78)	(1.09)	(3.25)	(0.48)	(1.91)	(−0.21)	(−0.58)	(−2.14)
RET	1.0885***	0.5117***	0.8192***	0.4245***	0.3202***	−0.2802	−0.0290***	−0.5307***
	(5.07)	(4.45)	(4.38)	(4.94)	(4.78)	(−1.34)	(−5.96)	(−8.05)
SIGMA	–	–	–	0.0803***	−0.0149	0.2276***	0.0107***	0.1087***
				(5.06)	(−1.33)	(10.37)	(15.86)	(7.18)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.3953	0.3289	0.3777	0.0889	0.0839	0.6226	0.8195	0.6946
Obs.	9213	9213	9213	9213	9213	9213	9213	9213

Panel B. ESG_S

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	SIGMA	BETA	IVOL_FF3	NCSKEW	DUVOL	KZ_Index	WW_Index	SA_Index
Intercept	2.7822***	−0.0610	2.8899***	−1.5633***	−0.8360***	−0.7994***	−0.3471***	−7.2502***
	(19.21)	(−0.54)	(22.72)	(−15.73)	(−11.85)	(−4.55)	(−58.37)	(−65.88)
ESG_S	−0.0040***	−0.0028***	−0.0032***	−0.0002	−0.0002	−0.0090***	−0.0002***	−0.0032***
	(−4.74)	(−4.14)	(−4.33)	(−0.22)	(−0.35)	(−4.24)	(−5.66)	(−2.94)
LEV	0.3432***	0.0895***	0.3146***	0.0436	−0.0142	4.3446***	−0.0566***	0.7266***
	(7.26)	(3.22)	(6.55)	(0.99)	(−0.43)	(55.10)	(−19.05)	(19.78)
ROA	−3.6835***	−0.0648	−3.6473***	−0.1350	−0.0982	−10.7800**	−0.1750***	−0.1382
	(−21.51)	(−0.93)	(−21.62)	(−0.99)	(−0.90)	(−38.75)	(−18.64)	(−0.98)
MB	0.1348***	−0.0217***	0.1521***	0.0631***	0.0546***	−0.2894***	0.0130***	−0.1370***
	(11.54)	(−5.83)	(15.41)	(8.81)	(9.30)	(−15.94)	(30.04)	(−17.71)
Ln(MKTCAP)	−0.1145***	0.1001***	−0.1533***	0.0460***	0.0127*	0.0499***	−0.0475***	0.6048***
	(−10.50)	(10.82)	(−14.17)	(4.63)	(1.83)	(4.80)	(−100.14)	(48.94)
DTURNOVER	0.0631***	0.0203	0.0534***	0.0072	0.0243*	−0.0058	−0.0005	−0.0247**
	(2.76)	(1.08)	(3.24)	(0.48)	(1.91)	(−0.16)	(−0.51)	(−2.14)
RET	1.0870***	0.5103***	0.8182***	0.4247***	0.3204***	−0.2706	−0.0288***	−0.5269***
	(5.11)	(4.52)	(4.41)	(4.94)	(4.78)	(−1.30)	(−5.86)	(−8.11)
SIGMA	–	–	–	0.0804***	−0.0151	0.2192***	0.0105***	0.1070***
				(5.03)	(−1.35)	(9.78)	(15.79)	(7.37)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.3964	0.3318	0.3786	0.0889	0.0839	0.6239	0.8198	0.6941
Obs.	9213	9213	9213	9213	9213	9213	9213	9213

Panel C. ESG_G

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	SIGMA	BETA	IVOL_FF3	NCSKEW	DUVOL	KZ_Index	WW_Index	SA_Index
Intercept	2.8317***	−0.0810	2.9427***	−1.6154***	−0.8783***	−0.7785***	−0.3479***	−7.6415***
	(17.75)	(−0.65)	(20.17)	(−16.02)	(−13.02)	(−5.04)	(−53.71)	(−96.87)
ESG_G	−0.0033***	−0.0012***	−0.0030***	0.0009	0.0007*	−0.0058***	−0.0001***	0.0059***

(*continued on next page*)

Table 3 (*continued*)

Panel C. ESG_G								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	SIGMA	BETA	IVOL_FF3	NCSKEW	DUVOL	KZ_Index	WW_Index	SA_Index
	(−4.00)	(−2.68)	(−3.77)	(1.63)	(1.96)	(−5.49)	(−2.93)	(9.55)
LEV	0.3328***	0.0867***	0.3052***	0.0467	−0.0118	4.3265***	−0.0570***	0.7465***
	(7.06)	(3.26)	(6.33)	(1.07)	(−0.36)	(55.49)	(−20.14)	(20.61)
ROA	−3.6121***	−0.0409	−3.5830***	−0.1521	−0.1116	−10.6475**	−0.1721***	−0.2410*
	(−22.23)	(−0.64)	(−22.45)	(−1.11)	(−1.03)	(−38.55)	(−18.08)	(−1.79)
MB	0.1401***	−0.0184***	0.1565***	0.0627***	0.0543***	−0.2787***	0.0133***	−0.1381***
	(11.85)	(−5.04)	(15.61)	(8.68)	(9.12)	(−15.09)	(31.48)	(−16.79)
Ln(MKTCAP)	−0.1269***	0.0902***	−0.1632***	0.0445***	0.0114	0.0204**	−0.0483***	0.5875***
	(−12.69)	(11.17)	(−16.80)	(4.66)	(1.65)	(2.11)	(−114.91)	(39.29)
DTURNOVER	0.0645***	0.0209	0.0547***	0.0065	0.0238*	−0.0039	−0.0004	−0.0293**
	(2.91)	(1.12)	(3.42)	(0.44)	(1.89)	(−0.11)	(−0.49)	(−2.55)
RET	1.0668***	0.5042***	0.7998***	0.4293***	0.3239***	−0.3093	−0.0296***	−0.5003***
	(5.03)	(4.44)	(4.32)	(4.99)	(4.85)	(−1.48)	(−6.09)	(−7.61)
SIGMA	–	–	–	0.0817***	−0.0140	0.2226***	0.0106***	0.1181***
				(5.15)	(−1.24)	(10.01)	(15.53)	(7.63)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.3965	0.3295	0.3788	0.0890	0.0842	0.6233	0.8195	0.6974
Obs.	9213	9213	9213	9213	9213	9213	9213	9213

This table presents the regression analysis on the effects of environmental, social, and governance scores on firm risks from 2015 to 2021 with the employment of cluster standard errors. Panels A, B, and C present the results for environmental, social, and governance scores. For financial risks, columns (1), (2), and (3) present the results on total risk, systematic risk, and idiosyncratic risk, respectively. For stock price crash risks, columns (4) and (5) present the results on the negative coefficient of skewness and down-to-up volatility, respectively. For financial constraints, columns (6), (7), and (8) present the results on KZ, WW, and SA indices, respectively. The independent variables are the values from the previous year (*t-1*). Year and industry fixed effects are included. Numbers in parentheses are *t*-values. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels.

4.3. ESG, firm risks, and the COVID-19 pandemic

To examine whether ESG engagement offers firms additional protection during periods of extreme market stress, we investigate the relationship between ESG performance and firm risks during the COVID-19 pandemic. Specifically, we conduct regressions with the fixed effects as follows.

$$\begin{aligned} Firm\ risks_{i,t} = {} & \alpha + \beta_1 ESG_{i,t-1} + \beta_2 ESG_{i,t-1} \times Year_{2020} + \beta_3 ESG_{i,t-1} \times Year_{2021} + \beta_4 LEV_{i,t-1} + \beta_5 ROA_{i,t-1} + \beta_6 MB_{i,t-1} \\ & + \beta_7 Ln(MKTCAP)_{i,t-1} + \beta_8 DTURNOVER_{i,t-1} + \beta_9 RET_{i,t-1} + \beta_{10} SIGMA_{i,t-1} + Industry_Dummies + Year_Dummies + \varepsilon_{i,t} \end{aligned} \quad (2)$$

where the dependent variable is one of the firm risk measures, financial risks, crash risks, and financial constraints, for firm *i* in year *t*. The model includes the same independent and control variables as specified in Eq. (1), with additional interaction terms between the ESG score and year dummies for 2020 and 2021, which correspond to the peak periods of the COVID-19 impact on Taiwan's capital market.

Table 4 presents the empirical results for the effect of ESG on firm risks during the COVID-19 period. Consistent evidence appears in columns (2) and (3), where systematic risk and idiosyncratic risk are the dependent variables. The coefficients on the interaction terms between ESG and the 2020 and 2021 dummies are significantly negative, indicating that ESG-performing firms experienced lower levels of both market-wide and firm-specific risk during the pandemic. These results further confirm that ESG practices enhance a firm's resilience, particularly under adverse market conditions.

Columns (4) and (5) show significantly negative coefficients for the interaction between ESG and the 2020 dummy. This suggests that during the COVID-19 crisis, ESG performance played a meaningful role in mitigating downside risk. This finding contrasts with the results in normal periods (Section 4.1), where no significant association is observed, implying that ESG engagement may function as a form of crisis insurance, becoming more salient when firms are exposed to heightened market volatility and uncertainty.

4.4. Heterogeneity differences

To explore how firm characteristics influence the relationship between ESG performance and firm risks, we divide the sample into two groups: large firms and small firms, using the median market capitalization as the cutoff. We then re-estimate the regression models as indicated by Eq. (1) for each subsample to determine whether the risk-mitigating effects of ESG differ by firm size.

Panel A of Table 5 focuses on financial risks and shows a significantly negative association between ESG scores and financial risk measures for large firms. The negative effects are more pronounced in both magnitude and statistical significance compared to those

Table 4
The effect of ESG on firm risks during the COVID-19 crisis.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	SIGMA	BETA	IVOL_FF3	NCSKEW	DUVOL	KZ_Index	WW_Index	SA_Index
Intercept	2.8502***	−0.1571	2.9463***	−1.6905***	−0.8898***	−0.4973**	−0.3428***	−7.3974***
	(18.20)	(−1.33)	(20.87)	(−17.01)	(−11.86)	(−2.39)	(−50.57)	(−69.17)
ESG	−0.0054***	−0.0007	−0.0046***	0.0023	0.0010	−0.0144***	−0.0003***	0.0012
	(−4.42)	(−0.79)	(−3.83)	(1.62)	(1.17)	(−5.66)	(−4.62)	(0.94)
ESG × Year_2020	−0.0033	−0.0036***	−0.0055**	−0.0061***	−0.0022*	0.0015	−0.0002	−0.0009
	(−1.17)	(−3.03)	(−2.01)	(−2.88)	(−1.91)	(0.31)	(−1.29)	(−0.20)
ESG × Year_2021	−0.0015	−0.0076***	0.0003	−0.0030	−0.0001	0.0024	0.0001	0.0001
	(−0.50)	(−4.83)	(0.12)	(−1.57)	(−0.12)	(0.46)	(0.53)	(0.01)
LEV	0.3376***	0.0887***	0.3095***	0.0450	−0.0135	4.3325***	−0.0569***	0.7281***
	(7.20)	(3.31)	(6.46)	(1.02)	(−0.41)	(54.72)	(−19.52)	(19.90)
ROA	−3.6240***	−0.0357	−3.5973***	−0.1409	−0.1031	−10.6600**	−0.1723***	−0.1384
	(−21.37)	(−0.52)	(−21.58)	(−1.03)	(−0.95)	(−38.21)	(−18.20)	(−0.96)
MB	0.1362***	−0.0204***	0.1530***	0.0631***	0.0548***	−0.2849***	0.0131***	−0.1343***
	(11.35)	(−5.50)	(15.22)	(8.78)	(9.40)	(−15.74)	(31.18)	(−15.88)
Ln(MKTCAP)	−0.1134***	0.0974***	−0.1513***	0.0445***	0.0107	0.0492***	−0.0476***	0.5900***
	(−10.17)	(12.15)	(−14.61)	(4.66)	(1.54)	(4.86)	(−117.11)	(42.37)
DTURNOVER	0.0641***	0.0215	0.0543***	0.0075	0.0242*	−0.0037	−0.0004	−0.0257**
	(2.87)	(1.17)	(3.35)	(0.50)	(1.91)	(−0.10)	(−0.46)	(−2.21)
RET	1.0706***	0.5014***	0.8044***	0.4252***	0.3212***	−0.2995	−0.0294***	−0.5285***
	(5.03)	(4.45)	(4.32)	(4.99)	(4.82)	(−1.45)	(−6.03)	(−7.94)
SIGMA	–	–	–	0.0811***	−0.0143	0.2156***	0.0105***	0.1119***
				(5.09)	(−1.27)	(9.69)	(15.38)	(7.52)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.3971	0.3336	0.3796	0.0894	0.0840	0.6243	0.8199	0.6932
Obs.	9213	9213	9213	9213	9213	9213	9213	9213

This table presents the regression analysis on the effect of ESG score on firm risks during the COVID waves of 2020 to 2021 with the employment of cluster standard errors. For financial risks, columns (1), (2), and (3) present the results on total risk, systematic risk, and idiosyncratic risk, respectively. For stock price crash risks, columns (4) and (5) present the results on the negative coefficient of skewness and down-to-up volatility, respectively. For financial constraints, columns (6), (7), and (8) present the results on KZ, WW, and SA indices, respectively. The independent variables are the values from the previous year (*t-1*), except for Year_2020 and Year_2021. Year and industry fixed effects are included. Numbers in parentheses are *t*-values. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels.

observed for small firms. This indicates that larger firms benefit more consistently from ESG engagement in managing both firm-specific and market-wide risks.

Panel B shows that, for large firms, ESG scores are negatively associated with stock price crash risks. This suggests that ESG engagement helps mitigate crash risks in the group of large firms. In contrast, no significant relationships are found for small firms, reinforcing the fact that larger firms are better positioned to translate ESG practices into reputational and operational buffers during adverse market conditions.

Panel C presents the negative association between ESG scores and the KZ index in both subsamples. However, the negative relationship with the WW index is stronger for small firms, indicating that ESG practices may play a more prominent role in addressing information asymmetry and investor trust in these firms. For the SA index, ESG scores are insignificantly related to both subsamples, with coefficients in opposite directions. This result is consistent with earlier findings that the SA index is highly sensitive to firm size and structural characteristics.

Overall, the heterogeneity analysis reveals that the benefits of ESG engagement in reducing financial and crash risks are more pronounced for larger firms. This may reflect their greater resources, higher visibility, and stronger institutional pressure to implement and disclose ESG practices. For small firms, while ESG can also help alleviate certain financial constraints, its effects on risk reduction are more limited.

In addition to firm size, we also examine industry-level heterogeneity by dividing the sample into tech and non-tech firms. Tech firms are defined as those operating within the technology and communications industries. The results show that the ESG achievements of tech firms provide stronger protection against financial risks and financial constraints than those of non-tech firms. This may reflect the fact that many tech firms in Taiwan are part of the upstream supply chain, and they may commit more to ESG initiatives to meet the requirements of internationally recognized clients.[4]

[4] The detailed results of industry heterogeneity can be obtained from Table Online Appendix 1.

Table 5
The effect of ESG on firm risks in large and small firms.

Panel A. Financial risks

	(1)	(2)	(3)	(4)	(5)	(6)
	SIGMA		BETA		IVOL_FF3	
	Large	Small	Large	Small	Large	Small
Intercept	2.5471***	4.1415***	0.4144***	−0.8168***	2.3813***	4.6513***
	(16.57)	(18.45)	(2.81)	(−5.68)	(19.21)	(20.84)
ESG	−0.0083***	−0.0050***	−0.0044***	0.0003	−0.0069***	−0.0054***
	(−4.63)	(−3.11)	(−4.19)	(0.40)	(−4.99)	(−3.44)
LEV	0.0817	0.5567***	0.1278***	0.0966***	0.0289	0.5333***
	(0.90)	(8.47)	(3.14)	(3.50)	(0.33)	(8.34)
ROA	−3.0275***	−3.6435***	−0.3719*	−0.3799***	−2.8499***	−3.4723***
	(−8.05)	(−16.56)	(−1.96)	(−4.70)	(−8.42)	(−17.34)
MB	0.1168***	0.1035***	0.0062	−0.0627***	0.1240***	0.1314***
	(10.54)	(4.73)	(1.45)	(−8.98)	(12.58)	(7.09)
Ln(MKTCAP)	−0.0506***	−0.2937***	0.0459***	0.1917***	−0.0582***	−0.3841***
	(−3.80)	(−11.33)	(5.33)	(12.50)	(−4.83)	(−15.80)
DTURNOVER	0.0462*	0.0507	0.0231	0.0047	0.0391*	0.0402
	(1.68)	(1.58)	(1.17)	(0.25)	(1.82)	(1.57)
RET	1.5779***	0.8030***	0.6083***	0.4247***	1.2854***	0.5866***
	(5.12)	(4.45)	(3.94)	(4.42)	(4.86)	(3.72)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Year	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Industry	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.3988	0.4062	0.3487	0.3016	0.3285	0.4149
Obs.	4609	4604	4609	4604	4609	4604

Panel B. Stock price crash risks

	(1)	(2)	(3)	(4)
	NCSKEW		DUVOL	
	Large	Small	Large	Small
Intercept	−1.9376***	−0.4026**	−1.2993***	0.1099
	(−11.66)	(−2.22)	(−12.53)	(0.84)
ESG	−0.0026*	0.0027	−0.0016*	0.0016
	(−1.82)	(1.50)	(−1.79)	(1.45)
LEV	−0.0581	0.1363***	−0.0684	0.0377
	(−0.89)	(3.08)	(−1.60)	(1.09)
ROA	0.8115***	−0.2398*	0.5944***	−0.1086
	(4.16)	(−1.67)	(4.60)	(−1.12)
MB	0.0336***	0.0900***	0.0309***	0.0839***
	(4.75)	(5.99)	(5.77)	(7.94)
Ln(MKTCAP)	0.1279***	−0.1377***	0.0855***	−0.1271***
	(9.23)	(−6.99)	(9.45)	(−8.60)
DTURNOVER	0.0150	0.0106	0.0294***	0.0306*
	(1.36)	(0.47)	(3.40)	(1.68)
RET	0.4943***	0.5698***	0.3507***	0.4226***
	(5.75)	(5.34)	(5.00)	(4.63)
SIGMA	0.0294*	0.0623**	−0.0462***	−0.0349*
	(2.00)	(2.53)	(−4.64)	(−2.00)
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Cluster by Year	Yes	Yes	Yes	Yes
Cluster by Industry	Yes	Yes	Yes	Yes
Adj. R^2	0.1163	0.1076	0.1365	0.1054
Obs.	4609	4604	4609	4604

Panel C. Financial constraints						
	(1)	(2)	(3)	(4)	(5)	(6)
	KZ_Index		WW_Index		SA_Index	
	Large	Small	Large	Small	Large	Small
Intercept	−1.1145***	0.0719	−0.3352***	−0.3346***	−8.3990***	−6.4122***
	(−3.96)	(0.24)	(−52.32)	(−26.92)	(−51.48)	(−54.60)
ESG	−0.0138***	−0.0138***	−0.0001*	−0.0005***	−0.0011	0.0021
	(−4.99)	(−5.05)	(−1.95)	(−4.49)	(−0.85)	(1.62)
LEV	4.3937***	4.2178***	−0.0619***	−0.0515***	0.8831***	0.6258***
	(39.82)	(46.75)	(−16.30)	(−14.97)	(14.41)	(17.14)
ROA	−13.3167**	−8.7857***	−0.1276***	−0.2048***	−0.0328	0.1731
	(−22.23)	(−25.16)	(−8.93)	(−19.25)	(−0.13)	(1.66)
MB	−0.2491***	−0.2586***	0.0127***	0.0118***	−0.1418***	−0.1065***
	(−11.97)	(−11.74)	(30.49)	(12.24)	(−14.54)	(−14.99)
Ln(MKTCAP)	0.0832***	−0.0355	−0.0488***	−0.0472***	0.6971***	0.4733***
	(3.82)	(−1.22)	(−111.71)	(−42.46)	(38.35)	(29.27)
DTURNOVER	0.0009	0.0100	−0.0014	0.0007	−0.0231	−0.0180
	(0.02)	(0.31)	(−1.12)	(0.56)	(−1.48)	(−1.42)
RET	−1.0527***	0.1525	−0.0350***	−0.0198***	−0.4985***	−0.5387***
	(−3.69)	(0.75)	(−5.20)	(−2.80)	(−4.50)	(−7.04)
SIGMA	0.3433***	0.1564***	0.0106***	0.0099***	0.1301***	0.0692***
	(10.38)	(7.00)	(11.52)	(10.47)	(5.47)	(5.41)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Year	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Industry	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.6512	0.5760	0.7656	0.6100	0.6610	0.3290
Obs.	4609	4604	4609	4604	4609	4604

This table presents the regression analysis on the effect of ESG score on firm risks in large and small firms from 2015 to 2021 with the employment of cluster standard errors. Large (small) firms are defined as those whose MKTCAP is above (below) the median value in a given year. Panel A presents the results for financial risks, including total risk, systematic risk, and idiosyncratic risk. Panel B presents the results for stock price crash risks, including the negative coefficient of skewness and down-to-up volatility. Panel C presents the results for financial constraints using KZ, WW, and SA indices. The independent variables are the values from the previous year (*t-1*). Year and industry fixed effects are included. Numbers in parentheses are *t*-values. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels.

5. Robustness check

5.1. Alternative ESG measures

To test the robustness of our findings, we employ an alternative measure of ESG performance based on an industry-specific framework developed by the Sustainability Accounting Standards Board (SASB). TEJ applies SASB's Sustainable Industry Classification System (SICS) to categorize Taiwan's firms into 77 industries across 11 thematic sectors, reflecting common sustainability challenges within each sector (see Sustainability Accounting Standards Board, 2018; Eccles and Klimenko, 2019; Henriksson et al., 2019). Using the SASB-based ESG scores provided by TEJ, we re-examine the relationship between a firm's ESG engagement and its risks. The empirical results, consistent with our baseline findings, show that firms with higher SASB-based ESG scores exhibit significantly lower financial risks and reduced financial constraints.

5.2. Risk decomposition by different asset pricing models

To ensure that our findings are not sensitive to the choice of asset pricing model, we conduct additional robustness checks using alternative frameworks to estimate idiosyncratic risk. In addition to Fama and French's (1993) three-factor model employed in our baseline analysis, we replicate the risk decomposition using three alternative models: the capital asset pricing model (CAPM), Carhart's (1997) four-factor model, and Fama and French's (2015) five-factor model. Across all model specifications, the results consistently show that firms with stronger ESG performance tend to exhibit lower levels of idiosyncratic risk. Therefore, the conclusion that ESG engagement contributes to reducing firm-specific uncertainty remains valid regardless of the underlying asset pricing framework.

5.3. Endogeneity

We employ various approaches to address potential endogeneity concerns. Specifically, to mitigate reverse causality, we use independent variables from the previous year (*t-1*) rather than contemporaneous values (*t*) in Eqs. (1) to (2) for the regression analysis. As shown in Tables 2 to 5, the results based on lagged ESG variables provide supportive and consistent evidence that ESG engagement precedes changes in firm risk.

Table 6
The effect of ESG on firm risks controlling for delayed information diffusion.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	SIGMA	BETA	IVOL_FF3	NCSKEW	DUVOL	KZ_Index	WW_Index	SA_Index
Intercept	2.7122***	−0.1133	2.9212***	−1.6612***	−0.9733***	−0.4314**	−0.3311***	−7.3042***
	(15.61)	(−0.98)	(19.33)	(−14.94)	(−12.92)	(−2.40)	(−66.44)	(−60.07)
ESG_{t-1}	−0.0068**	−0.0025	−0.0065***	−0.0025	−0.0009	−0.0052	−0.0004***	0.0011
	(−2.59)	(−1.58)	(−2.81)	(−1.28)	(−0.65)	(−1.28)	(−3.00)	(0.41)
ESG_{t-2}	−0.0002	−0.0010	0.0005	0.0033	0.0019	−0.0080*	0.0001	−0.0005
	(−0.06)	(−0.57)	(0.18)	(1.42)	(1.23)	(−1.95)	(0.77)	(−0.21)
LEV_{t-1}	0.3104***	0.0959***	0.2794***	0.0305	−0.0246	4.3033***	−0.0560***	0.7186***
	(6.09)	(3.55)	(5.51)	(0.66)	(−0.73)	(47.51)	(−17.53)	(18.37)
ROA_{t-1}	−3.6486***	−0.0396	−3.6171***	−0.1536	−0.0930	−10.7643**	−0.1660***	−0.1600
	(−18.92)	(−0.52)	(−18.98)	(−1.03)	(−0.80)	(−34.85)	(−16.64)	(−1.01)
MB_{t-1}	0.1290***	−0.0217***	0.1478***	0.0618***	0.0532***	−0.2843***	0.0129***	−0.1321***
	(9.76)	(−5.47)	(13.08)	(7.85)	(8.28)	(−13.84)	(30.88)	(−13.91)
$Ln(MKTCAP)_{t-1}$	−0.1054***	0.0958***	−0.1443***	0.0439***	0.0108	0.0442***	−0.0479***	0.5907***
	(−8.80)	(9.80)	(−12.57)	(4.06)	(1.39)	(4.13)	(−118.75)	(38.57)
$DTURNOVER_{t-1}$	0.0597**	0.0230	0.0493***	0.0110	0.0255*	−0.0197	−0.0008*	−0.0250*
	(2.47)	(1.14)	(2.82)	(0.68)	(1.85)	(−0.51)	(−1.68)	(−1.88)
RET_{t-1}	1.2395***	0.5467***	0.9531***	0.3598***	0.2647***	−0.2806	−0.0336***	−0.5215***
	(5.60)	(4.39)	(4.86)	(3.95)	(3.70)	(−1.19)	(−5.92)	(−7.02)
$SIGMA_{t-1}$	–	–	–	0.0718***	−0.0184	0.2337***	0.0106***	0.1090***
				(4.22)	(−1.48)	(10.48)	(13.33)	(6.68)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.4045	0.3453	0.3803	0.0856	0.0841	0.6152	0.8214	0.6895
Obs.	7776	7776	7776	7776	7776	7776	7776	7776

This table presents the regression analysis on the effect of ESG score on firm risks from 2015 to 2021 with the employment of cluster standard errors and the control of ESG score from the year before last (*t-2*). For financial risks, columns (1), (2), and (3) present the results on total risk, systematic risk, and idiosyncratic risk, respectively. For stock price crash risks, columns (4) and (5) present the results on the negative coefficient of skewness and down-to-up volatility, respectively. For financial constraints, columns (6), (7), and (8) present the results on KZ, WW, and SA indices, respectively. The independent variables are the values from the previous year (*t-1*), except for the ESG score from the year before last (*t-2*). Year and industry fixed effects are included. Numbers in parentheses are *t*-values. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels.

Following the approach of Feng et al. (2022), we test for delayed information transmission by incorporating the two-year lagged ESG score as an additional explanatory variable. Table 6 shows that one-year lagged ESG scores remain significantly negatively associated with risk measures, while two-year lagged ESG scores are largely insignificant. This suggests that the effect of ESG on firm risks is more immediate and that concerns about delayed information diffusion are limited.

To mitigate both reverse causality and omitted variable biases, we conduct the two-stage least squares (2SLS) analysis. Given that there is less concern about the delayed information diffusion in Table 6, we use a two-year lagged ESG as an instrumental variable for a one-year lagged ESG in the first-stage regression. Then, in the second-stage regression, we use the predicted one-year lagged ESG to estimate its effects on firm risks. In Panels A and B of Table 7, column (1) shows that the coefficient for two-year lagged ESG is significantly positive for one-year lagged ESG in the first-stage regression. Moreover, the second-stage regressions in Panel A show that the coefficients for predicted one-year lagged ESG are significantly negative for all of SIGMA, BETA, and IVOL_FF3. In Panel B, the coefficients for predicted one-year lagged ESG are significantly negative for KZ_Index and WW_Index, while they remain insignificant for NCSKEW, DUVOL, and SA_Index. These findings are consistent with our baseline results and confirm the robustness of the risk-reducing effect of ESG engagement, even after addressing potential endogeneity.[5,6]

6. Conclusion

This study investigates the association between ESG scores and various dimensions of firm risk in Taiwan's capital market, including financial risks, stock price crash risks, and financial constraints. This study shows a negative association between ESG scores and multiple types of firm risks, providing evidence that ESG engagement functions as a form of insurance-like protection in Taiwan. Additionally, this study examines the impact of ESG performance on firm risks during the COVID-19 crisis of 2020 and 2021, demonstrating that companies with stronger ESG performance experience lower levels of systematic and idiosyncratic risk while also

[5] To mitigate simultaneity and unobservable heterogeneity problems, we refer to Bae et al. (2021), El Ghoul et al. (2011), and Wintoki et al. (2012) and conduct dynamic panel GMM. In Table Online Appendix 2, the results are consistent with Table 2.

[6] To mitigate time-invariant firm-specific omitted variable bias, we control for the firm fixed effects (Feng et al., 2022) and report in Table Online Appendix 3. In this specification, we use only ESG as an explanatory variable and control for both year and firm fixed effects. The results are still held.

Table 7
The effect of ESG on firm risks using 2SLS.

Panel A. Financial risks

	(1)	(2)	(3)	(4)
	First-stage	Second-stage		
	ESG_{t-1}	SIGMA	BETA	IVOL_FF3
Intercept	1.7213***	2.4124***	−0.2052***	2.6733***
	(5.00)	(29.63)	(−5.08)	(35.45)
ESG_{t-1}	–	−0.0070***	−0.0035***	−0.0060***
		(−4.80)	(−4.98)	(−4.45)
ESG_{t-2}	0.9070***	–	–	–
	(164.88)			
LEV_{t-1}	0.1105	0.3109***	0.0959***	0.2799***
	(0.59)	(6.62)	(4.32)	(6.30)
ROA_{t-1}	2.7311***	−3.6389***	−0.0354	−3.6099***
	(5.17)	(−26.30)	(−0.53)	(−27.37)
MB_{t-1}	−0.0743***	0.1279***	−0.0218***	0.1467***
	(−2.70)	(17.08)	(−6.13)	(20.73)
$Ln(MKTCAP)_{t-1}$	0.4374***	−0.1053***	0.0961***	−0.1445***
	(13.02)	(−13.00)	(23.76)	(−18.92)
$DTURNOVER_{t-1}$	−0.0701	0.0573***	0.0218***	0.0475***
	(−1.17)	(4.35)	(2.69)	(3.96)
RET_{t-1}	0.2537	1.2522***	0.5500***	0.9644***
	(0.69)	(13.93)	(12.21)	(11.51)
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Adj. R^2	–	0.4064	0.3470	0.3823
Obs.	7776	7776	7776	7776

Panel B. Stock price crash risks and financial constraints

	(1)	(2)	(3)	(4)	(5)	(6)
	First-stage	Second-stage				
	ESG_{t-1}	NCSKEW	DUVOL	KZ_Index	WW_Index	SA_Index
Intercept	2.1609***	−1.5348***	−0.8670***	−0.5112***	−0.3454***	−7.3409***
	(5.75)	(−18.15)	(−15.96)	(−3.45)	(−83.02)	(−111.48)
ESG_{t-1}	–	0.0011	0.0013	−0.0140***	−0.0003***	0.0005
		(0.82)	(1.44)	(−6.05)	(−4.18)	(0.46)
ESG_{t-2}	0.9053***	–	–	–	–	–
	(163.57)					
LEV_{t-1}	0.1664	0.0300	−0.0249	4.3039***	−0.0560***	0.7184***
	(0.88)	(0.72)	(−0.90)	(58.11)	(−23.74)	(21.42)
ROA_{t-1}	2.3034***	−0.1626	−0.0992	−10.7139***	−0.1658***	−0.1618*
	(4.23)	(−1.48)	(−1.24)	(−40.01)	(−21.06)	(−1.77)
MB_{t-1}	−0.0470	0.0616***	0.0530***	−0.2846***	0.0129***	−0.1317***
	(−1.61)	(11.52)	(13.35)	(−18.41)	(28.22)	(−24.33)
$Ln(MKTCAP)_{t-1}$	0.4202***	0.0424***	0.0100**	0.0479***	−0.0479***	0.5904***
	(12.28)	(5.96)	(2.07)	(3.55)	(−127.27)	(93.33)
$DTURNOVER_{t-1}$	−0.0325	0.0103	0.0246***	−0.0180	−0.0007	−0.0245***
	(−0.54)	(1.02)	(3.30)	(−0.68)	(−1.09)	(−2.78)
RET_{t-1}	0.5065	0.3604***	0.2663***	−0.2882*	−0.0338***	−0.5208***
	(1.33)	(5.76)	(5.74)	(−1.70)	(−7.48)	(−8.74)
$SIGMA_{t-1}$	−0.1621***	0.0722***	−0.0181**	0.2322***	0.0106***	0.1091***
	(−2.88)	(5.77)	(−2.18)	(10.02)	(15.76)	(11.81)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	–	0.0879	0.0864	0.6158	0.8218	0.6907
Obs.	7776	7776	7776	7776	7776	7776

This table presents the regression analysis on the effect of ESG score on firm risks from 2015 to 2021, using the two-stage least squares (2SLS) approach and the ESG score from the year before last (*t-2*) as an instrumental variable. Panel A presents the results for financial risks, including total risk, systematic risk, and idiosyncratic risk. Panel B presents the results for stock price crash risks, including the negative coefficient of skewness and down-to-up volatility, as well as for financial constraints using KZ, WW, and SA indices. The independent variables are the values from the previous year (*t-1*), except for the ESG score from the year before last (*t-2*). Year and industry fixed effects are included. Numbers in parentheses are *t*-values. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels.

mitigating crash risks.

Overall, despite ongoing debates about the necessity of non-financial disclosures like ESG reporting, this study presents compelling evidence supporting the effectiveness of ESG engagement. In times of financial turbulence, public sustainability disclosures and corporate social responsibility practices indeed help firms navigate uncertainties. Although implementing ESG initiatives involves significant costs, which may appear contradictory to the principle of maximizing shareholder value, the long-term benefits of risk reduction and financial stability are clear. These findings underscore the extensive adoption of GRI standards by companies in Taiwan and strongly reinforce the commitment of Taiwanese businesses to advancing their ESG journey.

CRediT authorship contribution statement

Shun-Fen Chang: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Bai-Sian Chen:** Writing – review & editing, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hong-Yi Chen:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization. **Hsiao-Yin Chen:** Writing – review & editing.

Acknowledgements

Hong-Yi Chen gratefully acknowledges financial support from National Science and Technology Council in Taiwan for awarding the Grant 111-2410-H-004-185-MY3.

Appendix A. Variable definitions

Variable	Name	Definition
ESG	ESG score	An aggregate score reflecting corporate efforts on environment, society, and governance
ESG_E	Environmental score	A score reflecting corporate environmental efforts
ESG_S	Social score	A score reflecting corporate social efforts
ESG_G	Governance score	A score reflecting corporate governance efforts
SIGMA	Total risk	The standard deviation of daily stock returns over a year
BETA	Systematic risk	The Beta coefficient from the Capital Asset Pricing Model (CAPM) for a year using daily stock returns
IVOL_FF3	Idiosyncratic risk	The standard deviation of daily residuals from Fama and French's (1993) three-factor model over a year
NCSKEW	Negative coefficient of skewness	$NCSKEW_{i,t} = \frac{-\left[n(n-1)^{\frac{3}{2}} \times \sum R_{i,d}^3\right]}{(n-1)(n-2) \times \left(\sum R_{i,d}^2\right)^{\frac{3}{2}}}$ where $NCSKEW_{i,t}$ is the value of the negative coefficient of skewness for firm i in year t, n denotes the number of demeaned firm-specific daily returns in the year, and $R_{i,d}$ denotes the demeaned firm-specific daily return for firm i on day d.
DUVOL	Down-to-up volatility	$DUVOL_{i,t} = ln\left[\frac{(n_{Up}-1) \times \sum_{Down} R_{i,d}^2}{(n_{Down}-1) \times \sum_{Up} R_{i,d}^2}\right]$ where $DUVOL_{i,t}$ is the value of the down-to-up volatility for firm i in year t, n_{Up} (n_{Down}) denotes the number of up (down) days in the year, and $R_{i,d}$ denotes the demeaned firm-specific daily return for firm i on day d.
KZ_Index	Kaplan and Zingales's (1997) index	$KZ_Index_{i,t} = -1.002 \times CF_{i,t}/TA_{i,t-1} - 39.368 \times DIV_{i,t}/TA_{i,t-1} - 1.315 \times C_{i,t}/TA_{i,t-1} + 3.139 \times LEV_{i,t} + 0.283 \times TQ_{i,t}$ where $CF_{i,t}$ is the cash flow of firm i in year t; $TA_{i,t-1}$ is the total assets of firm i in year t-1; DIV represents the cash dividends for both common and preferred shares; C stands for cash and cash equivalents; LEV is the ratio of total debt to the sum of total debt and total equity; and TQ is the market value of equity plus total assets minus the book value of equity, divided by total assets.
WW_Index	Whited and Wu's (2006) index	$WW_Index_{i,t} = -0.091 \times CF_{i,t}/TA_{i,t} - 0.062 \times DIVPOS_{i,t} + 0.021 \times LTD_{i,t}/TA_{i,t} - 0.044 \times Ln(TA)_{i,t} + 0.102 \times ISG_{i,t} - 0.035 \times SG_{i,t}$ where $CF_{i,t}$ is the cash flow of firm i in year t; $TA_{i,t}$ is the total assets of firm i in year t; $DIVPOS$ is a dummy variable that equals 1 if a firm pays cash dividends, and 0 otherwise; LTD represents long-term debt; $Ln(TA)$ is the natural logarithm of total assets; and ISG (SG) stands for industry (firm) sales growth.
SA_Index	Size and Age index	$SA_Index_{i,t} = -0.737 \times Size_{i,t} + 0.043 \times Size_{i,t}^2 - 0.040 \times Age_{i,t}$ where $Size_{i,t}$ is the natural logarithm of total assets of firm i in year t, and Age denotes the number of years since the firm was established.
LEV	Leverage ratio	The ratio of total debt to the sum of total debt and total stockholder's equity, with total debt comprising both short-term and long-term debt
ROA	Return on asset	The ratio of earnings before extraordinary items to total assets
MB	Market-to-book ratio	The ratio of the market value of equity to the book value of equity
MKTCAP	Market capitalization	The stock price multiplied by the number of shares outstanding
DTURNOVER	Change in trading volume	The difference between the average daily turnover for the current year and that for the previous year, where turnover is defined as the ratio of trading volume to the number of shares outstanding
RET	Past return	The average daily returns over a year

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pacfin.2025.102819.

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